

One hour of video from a camera can easily take up to 30 GB (gigabyte) of storage space, so one would need to use image compression techniques to keep storage requirements at a minimum. However, the compression should not result in too much loss of clarity in the images, otherwise the very point of storing the images would be lost. What's more, these image compression techniques need to be evaluated with CCTV images under different real-life lighting and imaging conditions. Similarly, while a number of imaging technologies exist, it is still a challenge to determine the right solutions for camera placement and lighting.

The brain behind the eyes

Monitoring of CCTV images has traditionally been done by human observers, who often monitor multiple screens simultaneously. As the number of cameras increases, it will be necessary to increase the number of human observers as well. However, studies have shown that after the first 40 minutes, the alertness of the human observer falls drastically. Hence, automated solutions to recognise events that seem out of the ordinary, need to be developed.

Machine vision technology is maturing and today, a computer program can automatically detect stationary and non-stationary objects in an image. For example, people, luggage, trolleys, dustbins, kiosks and other objects in images can be automatically identified by a computer program. A computer program can even detect unattended luggage with reasonable accuracy—an important factor considering

The need is for an imaging technology that has good resolutions under different imaging conditions, and is inexpensive and robust enough to be used outdoors and indoors.



Output of an image processing program which has identified all the pieces of luggage in the frame

that bombs are often left in the form of luggage, transistors, etc. A computer can also detect someone carrying a suspicious-looking object. A human observer can be alerted to look at such images, verify that the system is correct and notify security personnel on the ground.

It is possible to match the faces and gaits of people with those of known suspects, as well as identify suspicious behaviour like nervous or jerky movements by individuals, which can be indicative of a person who is about to carry out an attack.

It is essential that security experts work with the engineers writing such programs to define the scenarios of interest. The computer program needs to be given examples of normal events so that it can report anything that does not fit into this pattern.

Start-ups for security

But while security in public places needs to be stepped up, such security should not be disruptive. At CST, the authorities rejected a plan to frisk every person entering the premises—it was impractical to frisk over three million passengers every day. Security has to be effective but at the same time it should be unobtrusive as far as possible. Just placing guards at entrances or

even frisking is not a solution in itself. We need to put technology to effective use and collect as much actionable data as possible to pre-empt terrorist strikes.

ASSOCHAM expects that the security business in India will become a Rs 500 billion industry in the next four years. Much of this money will go into deploying technologies that can effectively fill the security gaps.

This represents an opportunity for technopreneurs to help make our lives safer. The first challenge is to reduce the cost of hardware. Today, a night-vision camera costs Rs 10,000, while a camera for normal lighting costs Rs 5000. These costs need to be brought down through innovative use of photo equipment and semiconductors. The next challenge is to develop better machine-vision software. As it is not possible to monitor all images manually, machine-vision algorithms for automatic surveillance need to be developed and deployed to aid the human observers. As in all matters these days, terrorism needs to be tackled by a blend of human effort and technology. ●

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